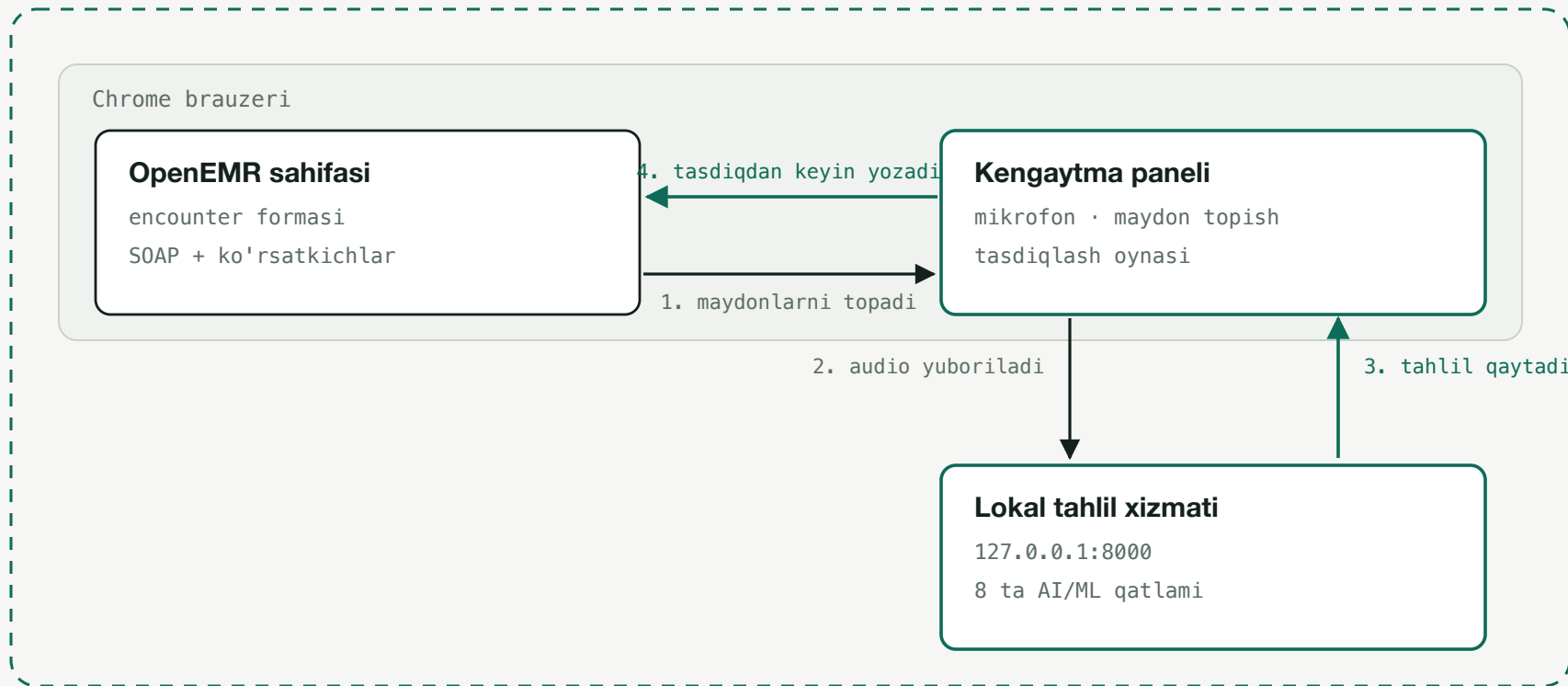


Arxitektura

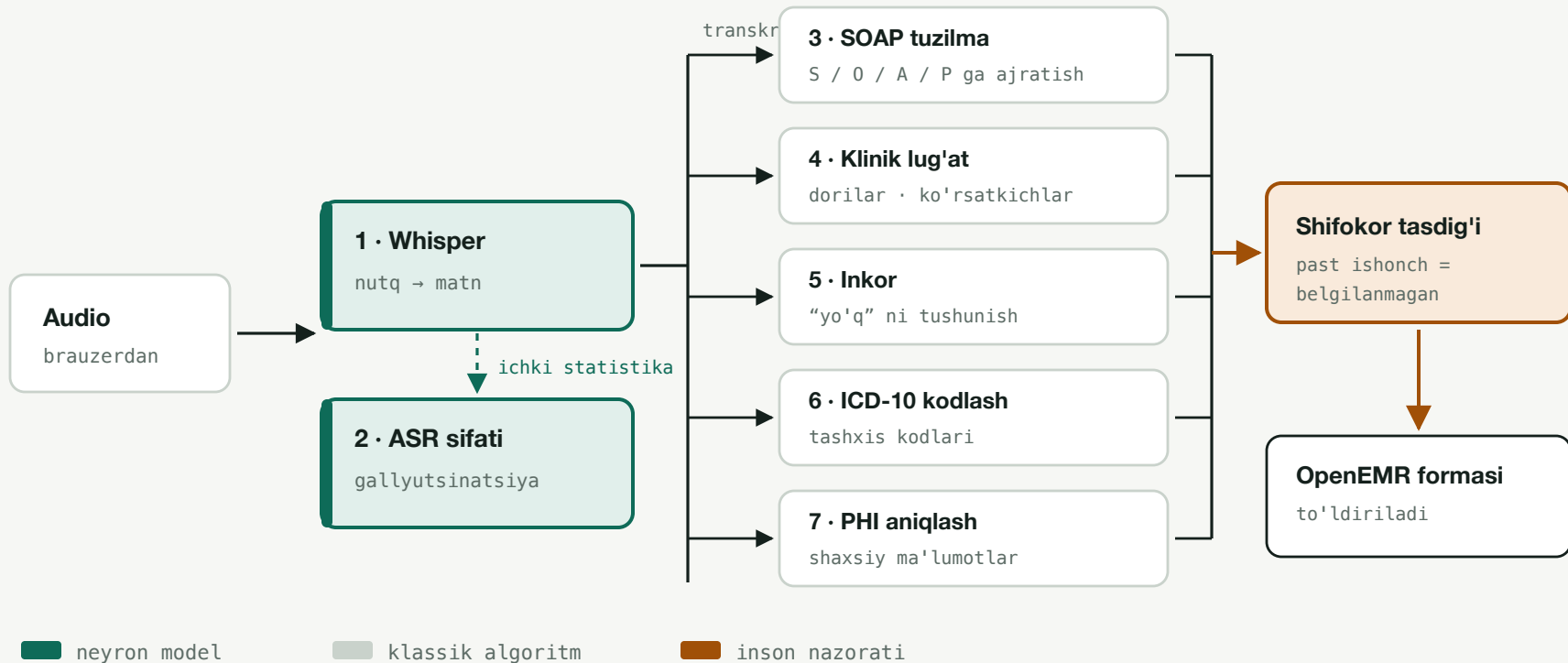
Uch qism, to'rt qadam — hammasi bitta kompyuter ichida

BITTA ISH STANTSIYASI – TARMOQQA HECH NARSA CHIQMAYDI



Sakkiz qatlam

Ikkitasi neyron tarmoq, beshtasi klassik algoritm, oxirida inson



Whisper to'qib chiqargan matnni qanday ushlaymiz

Bitta 16 soniyalik yozuv, to'rt segmentga bo'lingan

VAQT	TIZIM NIMA ESHITDI	ISHONCH	XULOSA
0:00 0:04	Patient reports headache and mild fever for two days. tabiiy nutq, hech qanday shubha yo'q	84% 	Yaxshi
0:04 0:07	Blood pressure was one forty over ninety. model o'zi ishonchsizligini bildirdi – qayta eshiting	26% 	Tekshiring
0:07 0:11	Thank you for watching, please subscribe. sukunat ehtimoli 0.88 – Whisper'ning hujjatlashtirilgan artefakti	80% 	TO'QILGAN
0:11 0:16	Follow up follow up follow up follow up follow up. ibora besh marta takrorlandi – bizning OpenEMR sinovimizda aynan shu chiqdi	58% 	TO'QILGAN

Diqqat: uchinchi qatorda ishonch 80% — yuqori. Lekin matn to'qilgan.

Faqat ishonch foiziga qarash yetarli emas — uchta mustaqil signal tekshiriladi. To'rtinchi qator bizning sinovimizdan.

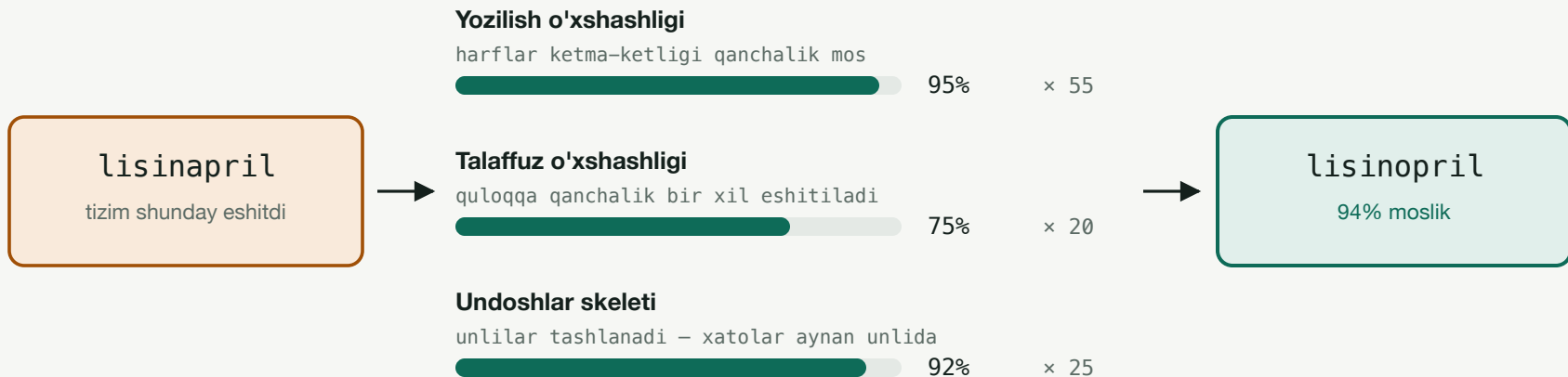
“Yo'q” degan so'zni tushunish

Bitta jumla, ikki xil natija



Noto'g'ri eshitilgan dori nomini tiklash

Uchta mustaqil o'lchov birlashtiriladi — hech biri yolg'iz yetarli emas



Chegara: butunlay buzilgan nomni hech qanday o'lchov tiklay olmaydi.

Haqiqiy sinovda “acetaminophen” → “a seed of minifin” bo'ldi. Bu yerda tibbiy model kerak.

Hozirgi holat va keyingi qadam

TO'LIQ ISHLAYDI

Ovoz → transkript → 8 qatlamli tahlil → shifokor tasdig'i → forma

45 ta avtomatik test · tahlil 31 millisekund · qo'shimcha kutubxonasiz ishlaydi

YAGONA OCHIQ NUQTA

Whisper — umumiy model

Dori nomlarini buzadi. Sinovda:

“acetaminophen” → “a seed of minifin”



Keyingi qadam: tibbiy ASR modeli

Bunday buzilishni satr algoritmi tiklay olmaydi —
bu yerda modelning o'zi almashishi kerak.